

Andrea Bonucci

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Education

- Università Bocconi**, MSc in Quantitative Finance and Risk Management Sep 2024 – Sep 2025
- **Coursework:** Stochastic Calculus for Finance, Derivatives Pricing, Structured Products, Fixed Income Securities, Market Risk, Credit Risk Management.
- Vrije Universiteit Amsterdam**, BSc in Mathematics Sep 2020 – Dec 2023
- **Thesis:** *Combinatorial Aspects of Surface Cluster Algebras and Applications to Frobenius' Conjecture* (Grade: 9/10).
 - **Honours:** Selected as one of three students to conduct independent research in advanced algebra.

Professional Experience

- Trading Systems Analyst**, FinIQ, London, UK Mar 2026 – Present
- Develop and maintain production pricing models for FX vanilla and exotic options (barriers, digitals, TARFs, accumulators) in FinIQ's hybrid pricing engine; construct volatility surfaces and implement stochastic and local volatility models.
 - Engineered an automated validation and stress-testing framework for an SLV pricing API: model-free no-arbitrage checks (put-call and in/out parity, Greek identities, finite-difference sensitivities), generated stress scenarios and per-RFQ audit reports.
 - Use $K(kdb+/q, K3, K6)$ for time-series modelling and analysis of pricing and market data.
 - Deliver quant sessions on pricing models for complex structured product payoffs and on random number generation for Monte Carlo simulation, to audiences of up to ~200.
 - Partner with sales and institutional clients on pricing and electronic execution of OTC structured products across FX and equity derivatives.
- Fixed Income Sales (Internship)**, Mediobanca, Milan, Italy Jul 2025 – Dec 2025
- Produced market reports, trade analysis and pricing support for institutional clients, working with the trading desk on market trends and liquidity.
- Software Developer**, marXact, Hilversum, Netherlands Sep 2023 – Apr 2024
- Developed real-time signal-processing software in Python for GNSS receivers, using tilt compensation to improve positioning accuracy in low-visibility environments.
 - Implemented an Extended Kalman Filter (EKF) for real-time state estimation in a non-linear system.

Technical Projects

- High-Frequency Backtesting Engine** | $K(K3, K6), kdb+/q$  GitHub
- Built a zero-loop, fully vectorised backtesting engine in pure $K(K3/K6)$ and ported it to $kdb+/q$ to leverage its C-optimised vector primitives.
 - Designed an $O(N)$ prefix-sum and boolean-masking pipeline for parameter grid searches, evaluating 8.1 billion matrix operations over 164,000+ data points.
 - Parallelised the workload across 10 threads with q 's *peach*, peaking at ≈ 2.34 billion ops/sec (3.4 s end to end).
- HFT Options Tickerplant** | $kdb+/q$  GitHub
- Engineered a distributed, fault-tolerant $kdb+$ tick architecture: tickerplant (logging and routing), in-memory RDB, on-disk HDB and a parallel unified gateway.
 - Built a simulated market data feed (20,000 updates/sec) to stress-test tickerplant stability and RDB-to-HDB end-of-day flushing.
 - Implemented an asynchronous, parallel gateway query handler that merges historical and real-time data in a single query.

Skills

- **Languages:** $K(K3, K6)$, $kdb+/q$, Python (NumPy, pandas, SciPy), C (C99), MATLAB, R, Mathematica.
- **Tools:** Git, Linux/Bash, $\LaTeX$.
- **Engineering:** Vectorised / array programming, parallel computing (SMP), time-series databases and tick architectures, performance optimisation, automated testing and model validation.
- **Quantitative Methods:** Stochastic calculus, Monte Carlo (variance reduction, quasi-Monte Carlo), finite-difference methods, volatility surface calibration (SVI, arbitrage-free), stochastic and local volatility, Kalman filtering, numerical methods.